WI4014TU – Numerical Analysis for PDE's

Lecture 6
Finite Volume Method in 1D, approximation order, conservative property

Finite Volume Method in 1D

Local conservation laws

We are specifically looking at equations of the type:

$$w' + qu = f$$

with and additional equation for w, e.g., $w = -ku'$. This is a one-dimensional local conservation law. In more dimensions it could be:

$$\nabla \cdot \boldsymbol{j} + qu = f$$

for example with $\boldsymbol{j} = -k\nabla u$.

Example: steady-state temperature distribution

Steady-state 1D problem (shielded Poisson's equation)

$$
\begin{aligned}
&-\left(ku'\right)' + qu = f\,, \quad x \in (a,b)\\
&-k(a)u'(a) + \beta u(a) = \mu_1, \quad u(b) = \mu_2
\end{aligned}
$$

describes the stationary distribution of temperature in a rod with one end ($x = b$) kept at temperature μ_2, while the other end ($x = a$) is exchanging heat with environment ($\beta \geq 0$).

$k(x) > 0$ – thermal conductivity coefficient
$q(x)u(x)$ – radiative heat loss term ($q(x) \geq 0$)
$f(x)$ – density of thermal sources

FVM vs FDM

Finite-Difference Method attempts to solve PDE's at grid nodes x_i, $i = 1, \dots, N-1$:

$$-(ku')'_i + qu_i = f_i$$

applying finite-difference approximation for the derivative $-(ku')'_i$.

Finite-Volume Method attempts to solve PDE's on average over sub-intervals $[a_i, b_i]$, $i = 1, \dots, N-1$; that divide the domain $\Omega = [a, b]$

$$\frac{1}{b_i - a_i} \int_{a_i}^{b_i} \left[-(ku')' + qu \right] \, dx = \frac{1}{b_i - a_i} \int_{a_i}^{b_i} f \, dx$$

applying a mix of approximations for integrals and derivatives.

Vertex-centered FVM

Connection between grid nodes and sub-intervals

We introduce a nonuniform grid $\{x_i, i = 0, \ldots, N\}$ with the steps $h_i = x_i - x_{i-1}$, $i = 1, \ldots, N$

In addition, we introduce intermediate grid points that in *vertex-centered* FVM serve as boundaries of sub-intervals (finite volumes)

$$x_{i-1/2} = x_i - 0.5h_i; \quad x_{i+1/2} = x_i + 0.5h_{i+1}$$

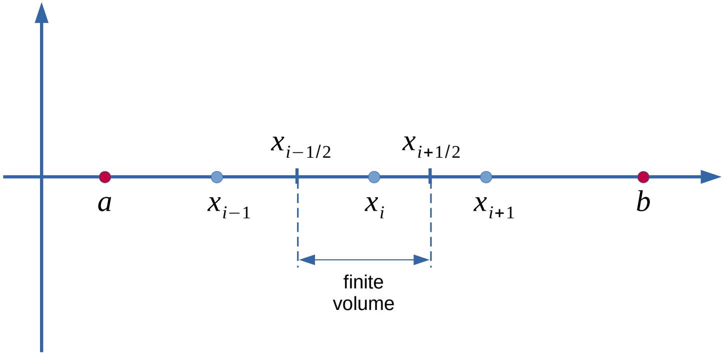
We also use the notation:

$$w(x) = -k(x)u'(x)$$

and for the grid values

$$u_i = u(x_i), \quad w_{i\pm1/2} = w(x_{i\pm1/2})$$

Staggered 1D grid with finite volumes



Common procedure (as in the book)

Step 1: integrate PDE over finite volumes

For all inner points x_i, $i = 1, \ldots, N-1$; integrate the equation over the "finite volume" $[x_{i-1/2}, x_{i+1/2}]$:

$$\int_{x_{i-1/2}}^{x_{i+1/2}} w'(x)\,dx + \int_{x_{i-1/2}}^{x_{i+1/2}} q(x)u(x)\,dx = \int_{x_{i-1/2}}^{x_{i+1/2}} f(x)\,dx$$

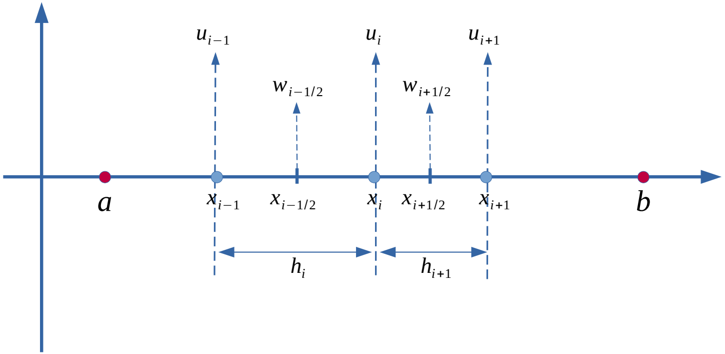
Step 2: use the FTC

Use the Fundamental Theorem of Calculus on the w' term:

$$w_{i+1/2} - w_{i-1/2} + \int_{x_{i-1/2}}^{x_{i+1/2}} q(x)u(x)\,dx = \int_{x_{i-1/2}}^{x_{i+1/2}} f(x)\,dx$$

No approximations have been made so far.

Locations of $w_{i\pm1/2}$ on the grid



Step 3: apply central-difference approximations for $w_{i\pm 1/2}$

Since $w_{i\pm 1/2} = -[ku']_{i\pm 1/2} = -k_{i\pm 1/2} u'_{i\pm 1/2}$, we need to approximate the first-order derivatives $u'_{i\pm 1/2}$.

We use $\mathcal{O}(h^2)$ central-difference approximations for $u'_{i\pm 1/2}$:

$$u'_{i+1/2} = \frac{u_{i+1} - u_i}{h_{i+1}} - \frac{u'''_{i+1/2}}{24} h_{i+1}^2 + \dots$$
$$u'_{i-1/2} = \frac{u_i - u_{i-1}}{h_i} - \frac{u'''_{i-1/2}}{24} h_i^2 + \dots$$

arriving at $\mathcal{O}(h^2)$ approximations for $w_{i\pm 1/2}$:

$$w_{i+1/2} = -k_{i+1/2} u'_{i+1/2} = -k_{i+1/2} \frac{u_{i+1} - u_i}{h_{i+1}} + \mathcal{O}(h_{i+1}^2)$$
$$w_{i-1/2} = -k_{i-1/2} u'_{i-1/2} = -k_{i-1/2} \frac{u_i - u_{i-1}}{h_i} + \mathcal{O}(h_i^2)$$

Step 4: approximate remaining integrals

Single-point approximations of the integrals:

$$\int_{x_{i-1/2}}^{x_{i+1/2}} q(x)u(x)\,dx \approx \frac{h_i + h_{i+1}}{2} q_i u_i$$

$$\int_{x_{i-1/2}}^{x_{i+1/2}} f(x)\,dx \approx \frac{h_i + h_{i+1}}{2} f_i$$

Inner point equation

$$-k_{i+1/2}\frac{u_{i+1}-u_i}{h_{i+1}}+k_{i-1/2}\frac{u_i-u_{i-1}}{h_i}+\frac{h_i+h_{i+1}}{2}q_iu_i\approx\frac{h_i+h_{i+1}}{2}f_i$$

Use this procedure to derive your FVM equations.

Some technical problems with standard derivation

What happens if $h_{i+1}, h_i \to 0$?

$$-k_{i+1/2}\frac{u_{i+1}-u_i}{h_{i+1}} + k_{i-1/2}\frac{u_i-u_{i-1}}{h_i} + \frac{h_i+h_{i+1}}{2}q_i u_i \approx \frac{h_i+h_{i+1}}{2}f_i$$

What is the order of approximation?

$$w_{i+1/2} - w_{i-1/2} = -k_{i+1/2}\frac{u_{i+1}-u_i}{h_{i+1}} + k_{i-1/2}\frac{u_i-u_{i-1}}{h_i} + \mathcal{O}(h_{i+1}^2) - \mathcal{O}(h_i^2)$$

What happens on a uniform grid $h_{i+1} = h_i = h$?
Note, $\mathcal{O}(h_{i+1}^2) - \mathcal{O}(h_i^2) \neq \mathcal{O}(h_{i+1}^2 - h_i^2)$.

Longer derivation: order of approximation

Averages over finite volumes

$$\frac{1}{x_{i+1/2} - x_{i-1/2}} \int_{x_{i-1/2}}^{x_{i+1/2}} w'(x)\, dx + \frac{1}{x_{i+1/2} - x_{i-1/2}} \int_{x_{i-1/2}}^{x_{i+1/2}} q(x) u(x)\, dx$$
$$= \frac{1}{x_{i+1/2} - x_{i-1/2}} \int_{x_{i-1/2}}^{x_{i+1/2}} f(x)\, dx$$

Single-point approximation of averaging integrals

Consider a generic integral of this type

$$\frac{2}{h_i + h_{i+1}} \int_{x_{i-1/2}}^{x_{i+1/2}} v(x)\, dx = \frac{2}{h_i + h_{i+1}} \int_{x_{i-1/2}}^{x_{i+1/2}} \left[v_i + v_i'(x - x_i) + \dots \right] \, dx$$

$$= \frac{2}{h_i + h_{i+1}} v_i \int_{x_{i-1/2}}^{x_{i+1/2}} \, dx + \frac{2}{h_i + h_{i+1}} v_i' \int_{x_{i-1/2}}^{x_{i+1/2}} (x - x_i)\, dx + \dots$$

$$= v_i + \frac{v_i'}{4} (h_{i+1} - h_i) + \mathcal{O}\left(\frac{h_{i+1}^3 + h_i^3}{h_{i+1} + h_i} \right)$$

Order of approximation (1)

In our case

$$\frac{2}{h_i + h_{i+1}} \int_{x_{i-1/2}}^{x_{i+1/2}} w'(x)\,dx = w_i' + \frac{w_i''}{4}(h_{i+1} - h_i) + \mathcal{O}\left(\frac{h_{i+1}^3 + h_i^3}{h_{i+1} + h_i}\right)$$

With $w' = -(ku')'$ we have

$$\frac{2}{h_i + h_{i+1}} \int_{x_{i-1/2}}^{x_{i+1/2}} w'(x)\,dx = -(ku')_i' + \frac{w_i''}{4}(h_{i+1} - h_i) + \mathcal{O}\left(\frac{h_{i+1}^3 + h_i^3}{h_{i+1} + h_i}\right)$$

Any $\mathcal{O}(h_{i+1} - h_i)$ FD approximation for $-(ku')_i'$ will lead to the total approximation of $\mathcal{O}(h_{i+1} - h_i)$.

Order of approximation (2)

For example, with $k = 1$ we have, $w' = -u''$. From the Taylor expansion one can derive:

$$u_i'' = \frac{2}{h_i + h_{i+1}} \left(\frac{u_{i+1} - u_i}{h_{i+1}} - \frac{u_i - u_{i-1}}{h_i} \right) + \frac{u_i'''}{3}(h_{i+1} - h_i) + \mathcal{O}\left(\frac{h_{i+1}^3 + h_i^3}{h_{i+1} + h_i} \right)$$

Hence

$$\frac{2}{h_i + h_{i+1}} \int_{x_{i-1/2}}^{x_{i+1/2}} [-u''(x)]\, dx = -\frac{2}{h_i + h_{i+1}} \left(\frac{u_{i+1} - u_i}{h_{i+1}} - \frac{u_i - u_{i-1}}{h_i} \right) + \mathcal{O}(h_{i+1} - h_i) + \mathcal{O}\left(\frac{h_{i+1}^3 + h_i^3}{h_{i+1} + h_i} \right)$$

Order of approximation (3)

With $w' = -(ku')'$, (some)one has to show that:

$$(ku')'_i = \frac{2}{h_i + h_{i+1}} \left(k_{i+1/2} \frac{u_{i+1} - u_i}{h_{i+1}} - k_{i-1/2} \frac{u_i - u_{i-1}}{h_i} \right) + \mathcal{O}(h_{i+1} - h_i)$$
$$+ \mathcal{O}\left(\frac{h_{i+1}^3 + h_i^3}{h_{i+1} + h_i} \right)$$

In which case it would follow that

$$\frac{2}{h_i + h_{i+1}} \int_{x_{i-1/2}}^{x_{i+1/2}} [-(ku')'(x)]\, dx =$$
$$- \frac{2}{h_i + h_{i+1}} \left(k_{i+1/2} \frac{u_{i+1} - u_i}{h_{i+1}} - k_{i-1/2} \frac{u_i - u_{i-1}}{h_i} \right)$$
$$+ \mathcal{O}(h_{i+1} - h_i) + \mathcal{O}\left(\frac{h_{i+1}^3 + h_i^3}{h_{i+1} + h_i} \right)$$

Approximating remaining integrals

We have two more integrals in our equation that need approximating:

$$\frac{2}{h_i + h_{i+1}} \int_{x_{i-1/2}}^{x_{i+1/2}} q(x)u(x)\,dx = q_i u_i + \mathcal{O}(h_{i+1} - h_i) + \mathcal{O}\left(\frac{h_{i+1}^3 + h_i^3}{h_{i+1} + h_i}\right)$$

$$\frac{2}{h_i + h_{i+1}} \int_{x_{i-1/2}}^{x_{i+1/2}} f(x)\,dx = f_i + \mathcal{O}(h_{i+1} - h_i) + \mathcal{O}\left(\frac{h_{i+1}^3 + h_i^3}{h_{i+1} + h_i}\right)$$

Inner-point equation

Applying all approximations we get

$$
-\frac{2}{h_i + h_{i+1}} \left(k_{i+1/2} \frac{u_{i+1} - u_i}{h_{i+1}} - k_{i-1/2} \frac{u_i - u_{i-1}}{h_i} \right) + q_i u_i = f_i
$$
$$
+ \mathcal{O}(h_{i+1} - h_i) + \mathcal{O}\left(\frac{h_{i+1}^3 + h_i^3}{h_{i+1} + h_i} \right)
$$

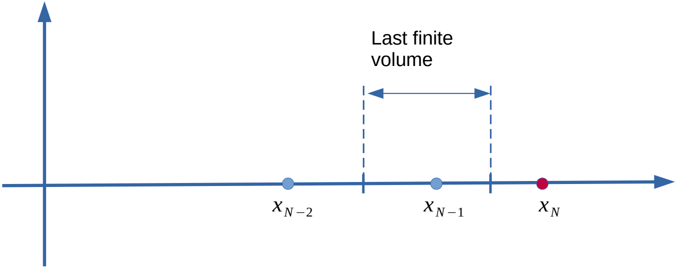
On the uniform grid $h_{i+1} = h_i = h$, the order of approximation becomes $\mathcal{O}(h^2)$.
If the grid is not very nonuniform, e.g., $h_{i+1} = h_i(1 - \mathcal{O}(h))$, where $h = \max_i h_i$, then the order of approximation is also $\mathcal{O}(h^2)$:

$$
\mathcal{O}(h_{i+1} - h_i) = \mathcal{O}\left(h_i(1 - \mathcal{O}(h)) - h_i\right) = \mathcal{O}(h_i \mathcal{O}(h)) = \mathcal{O}(h^2)
$$

Same is true for the FDM.

Boundary conditions

Grid near Dirichlet BC



Dirichlet BC

Use the inner point equation:

$$-\frac{2}{h_i + h_{i+1}}\left(k_{i+1/2}\frac{u_{i+1}-u_i}{h_{i+1}} - k_{i-1/2}\frac{u_i - u_{i-1}}{h_i}\right) + q_i u_i = f_i$$

$$+\mathcal{O}(h_{i+1}-h_i) + \mathcal{O}\left(\frac{h_{i+1}^3 + h_i^3}{h_{i+1}+h_i}\right)$$

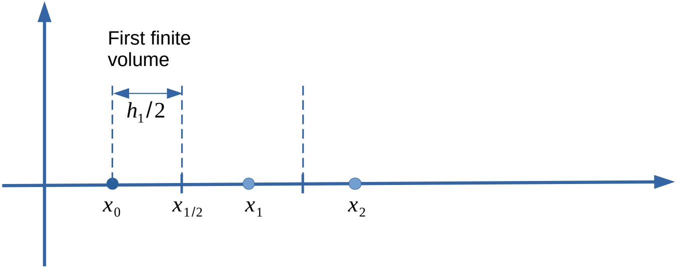
Vertex $i = N-1$:

$$-\frac{2}{h_{N-1} + h_N}\left(k_{N-1/2}\frac{u_N - u_{N-1}}{h_{N-1/2}} - k_{N-3/2}\frac{u_{N-1} - u_{N-2}}{h_{N-1}}\right) + q_{N-1} u_{N-1} = f_i$$

$$+\mathcal{O}(h_N - h_{N-1}) + \mathcal{O}\left(\frac{h_N^3 + h_{N-1}^3}{h_N + h_{N-1}}\right)$$

Set $u_N = \mu_2$ and shift to the right-hand side.
Order of approximation is the same as for other inner points.

Grid near Robin BC



Approximation of Robin BC

Robin BC: $-k(0)u'(0) + \beta u(0) = \mu_1$ or $w_0 + \beta u_0 = \mu_1$. It follows that:

$$w_0 = \mu_1 - \beta u_0$$

Average the equation over the first half-volume $[x_0, x_{1/2}]$

$$\frac{2}{h_1}\int_{x_0}^{x_{1/2}} w'(x)\,dx + \frac{2}{h_1}\int_{x_0}^{x_{1/2}} q(x)u(x)\,dx = \frac{2}{h_1}\int_{x_0}^{x_{1/2}} f(x)\,dx$$

Use FTC:

$$\frac{2}{h_1}(w_{1/2} - w_0) + \frac{2}{h_1}\int_{x_0}^{x_{1/2}} q(x)u(x)\,dx = \frac{2}{h_1}\int_{x_0}^{x_{1/2}} f(x)\,dx$$

Approximations at the Robin boundary

$$
\begin{aligned}
&\frac{2}{h_1}(w_{1/2} - w_0) = \frac{2}{h_1}\left(-k_{1/2}\frac{u_1 - u_0}{h_1} + \mathcal{O}(h_1^2) - \mu_1 + \beta u_0\right) \\
&= \frac{2}{h_1}\left(-k_{1/2}\frac{u_1 - u_0}{h_1} - \mu_1 + \beta u_0\right) + \mathcal{O}(h_1) \\
&\frac{2}{h_1}\int_{x_0}^{x_{1/2}} q(x)u(x)\,dx = q_0 u_0 + \mathcal{O}(h_1) \\
&\frac{2}{h_1}\int_{x_0}^{x_{1/2}} f(x)\,dx = f_0 + \mathcal{O}(h_1)
\end{aligned}
$$

Boundary equation for Robin BC

$$\frac{2}{h_1}\left(-k_{1/2}\frac{u_1 - u_0}{h_1} - \mu_1 + \beta u_0\right) + q_0 u_0 = f_0 + \mathcal{O}(h_1)$$

Shift what is known to the right-hand side.

Order of approximation has dropped to $\mathcal{O}(h_1)$. Does not help to have a uniform grid.

Conservative property of the FVM

Continuous conservation law

Integrating the main equation

$$-(ku')' = f - qu$$

over the whole domain $[a, b]$ we get the following conservation law:

$$w(b) - w(a) = \int_a^b [f(x) - q(x)u(x)]\, dx$$

where $w(x) = -k(x)u'(x)$.

Discrete conservation law

Analogously, adding up the discrete main equation for $i = 1, 2, \ldots, N-1$

$$-k_{i+1/2}\frac{u_{i+1}-u_i}{h_{i+1}} + k_{i-1/2}\frac{u_i-u_{i-1}}{h_i} + \frac{h_i+h_{i+1}}{2}q_i u_i = \frac{h_i+h_{i+1}}{2}f_i$$

we get the discrete version of the conservation law

$$-k_{N-1/2}\frac{u_N-u_{N-1}}{h_N} + k_{1/2}\frac{u_1-u_0}{h_1} = \sum_{i=1}^{N-1}\frac{h_i+h_{i+1}}{2}\left(f_i - q_i u_i\right)$$

Summary

- FVM is easy to derive (with standard procedure) for PDE's in the form of local conservation law
- Easy to work with nonuniform grids,
- The order of approximation is the same as with the FDM: $\mathcal{O}(h_{i+1} - h_i)$ on nonuniform and $\mathcal{O}(h^2)$ on uniform grids
- Boundary conditions involving fluxes (Neumann, Robin) are easy to handle, but introduce $\mathcal{O}(h)$ approximation, same as FDM
- Discrete conservative property is always satisfied with standard derivation procedure. Not always true with FDM
- Global error analysis is the same as with FDM: order of the global error is proportional to the order of the local approximation error. System matrix is difficult to analyze on nonuniform grids.